

Programs and Tools: Installations, Tips and Tricks

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Some instructions for installing and using some programs and tools.

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1 Beamer

Removing the headline and footline from the title page

Last modification: 2025-04-12

Tested with Beamer 3.72.

Use the `plain` option for removing the headline and footline areas from the title page:

```
\begin{frame}[plain]
\titlepage
\end{frame}
```

Removing the navigation symbols

Last modification: 2025-04-12

Tested with Beamer 3.72.

Write in the preamble the command

```
\setbeamertemplate{navigation symbols}{{}}
```

for suppressing all the navigation symbols.

2 Debian/GNU Linux on Windows via WSL

Last modification: 2025-04-12

For installing Debian on Windows we shall use the WSL (Windows Subsystem for Linux).

(i) Running PowerShell

For installing WSL open PowerShell in administrator mode by right-clicking and selecting *Run as administrator*:

```
Windows PowerShell  
C:\WINDOWS\system32>
```

(ii) Installing WSL

For installing WSL use the command

```
C:\WINDOWS\system32> wsl --install  
Downloading: Windows Subsystem for Linux 2.4.13  
Installing: Windows Subsystem for Linux 2.4.13  
...  
The operation completed successfully.
```

(iii) Available Linux distributions

To see a list of available Linux distributions available for installing with WSL use the command

```
C:\WINDOWS\system32> wsl --list --online
```

(iv) Installing Debian

For installing Debian run the commands

```
C:\WINDOWS\system32> wsl --install -d Debian
C:\WINDOWS\system32> wsl -d Debian
Enter new UNIX name:
New password:
Retype new password:
```

(v) Updating Debian

After installing Debian you should update it by running the commands:

```
$ sudo apt update
$ sudo apt upgrade
```

3 Emacs

Fixing trailing whitespace issues

Last modification: 2025-04-12

Tested with Emacs 29.4.

Add to the `.emacs` file the instruction `(setq-default_show-trailing-whitespace_t)` for showing whitespace and run the command `M-x delete-trailing-whitespace` for removing them.

4 Frama-C

Last modification: 2025-04-12

Tested with Framac-C 30.0

For installing the program Frama-C we assume you are using Debian or WSL/Debian (see Section 2).

First, run the following commands:

```
$ opam switch create frama-c-30.0 4.14.1  
$ eval $(opam env --switch=frama-c-30.0)  
$ opam install frama-c  
  
<> frama-c.30.0 installed successfully ><><><><><><><><><><><>  
=> The Frama-C/WP plug-in requires one or more external prover(s).  
Recommended provers are:  
 - Alt-Ergo (https://alt-ergo.ocamlpro.com)  
 - CVC4      (https://cvc4.github.io)
```

```
- CVC5      (https://cvc5.github.io)
- Z3        (https://github.com/Z3Prover/z3)
Use 'why3 config detect' to configure new provers.
```

=> Ivette is a new GUI for Frama-C, currently in development.
Run 'ivette' once to finalize installation (requires an internet connection).
Once finalized, 'ivette' will work offline.
Finalization also requires Node v20 or v22 and Yarn:

```
- install NVM (https://github.com/nvm-sh/nvm)
- run 'nvm use 22'
- run 'npm install --global yarn'
```

Now, run the following command:

```
$ why3 config detect
Prover Alt-Ergo version 2.6.0 is not recognized.
  Known versions for this prover: ^2\.5\.[2-9]$.
Prover Alt-Ergo (alternative: BV) version 2.6.0 is not recognized.
  Known versions for this prover: ^2\.5\.[2-9]$.
Prover Alt-Ergo (alternative: counterexamples) version 2.6.0 is not recognized.
  Known versions for this prover: ^2\.5\.[2-9]$.
Prover Alt-Ergo (alternative: FPA) version 2.6.0 is not recognized.
  Known versions for this prover: ^2\.4\.[0-3]$.
4 prover(s) added (including 4 prover(s) with an unrecognized version)
Save config to /home/asr/.why3.conf
```

5 GHC

Last modification: 2025-08-27

For installing GHC we suggest to use the `ghcup` tool (see Section 6). If after the installation you need a different version of GHC, you can also use the `ghcup` tool. For example,

- (i) Installing GHC 9.12.2

```
$ ghcup install ghc 9.12.2
```

- (ii) Switching to GHC 9.12.2

```
$ ghcup set ghc 9.12.2
```

- (iii) Testing the current version of GHC

```
$ ghc --version
The Glorious Glasgow Haskell Compilation System, version 9.12.2
```

6 ghcup

Last modification: 2025-08-27

For installing the `ghcup` tool we assume you are using Debian or WSL/Debian (see Section 2).

- (i) Before installing `ghcup` you need to install the required [Debian packages](#) for your version of Debian:

```
$ cat /etc/debian_version
12.10
```

```
$ sudo apt install build-essential curl libffi-dev libffi8 \
libgmp-dev libgmp10 libncurses-dev libncurses5 libtinfo5
```

- (ii) Now, [run the following command](#) for installing `ghcup` and chose the default options:

```
$ curl --proto '=https' --tlsv1.2 -sSf https://get-ghcup.haskell.org | sh
```

- (iii) Restart your terminal for the changes to take effect.

- (iv) Testing your installation by running the command

```
$ ghcup --version
The GHCup Haskell installer, version 0.1.50.1
```

- (v) Testing the GHC version installed by default.

```
$ ghc --version
The Glorious Glasgow Haskell Compilation System, version 9.6.7
```

7 Lambda Shell

Last modification: 2025-08-27

The Lambda Shell “is a feature-rich shell environment and command-line tool for evaluating terms of the pure, untyped lambda calculus” writing in Haskell.

The next instructions for installing the Lambda Shell program assume you are using Debian or WSL/Debian (see Section 2) and you installed GHC via the `ghcup` tool (see Section 6).

Instructions for installing the Lambda Shell program:

(i) Install GHC 9.12.2, if necessary (see Section 5).

(ii) Installing `git`, if necessary

```
$ sudo apt install git
```

(iii) Cloning the GitHub repository

```
$ git clone https://github.com/asr/lambda-shell.git  
$ cd lambda-shell
```

(iv) Installing Lambda Shell

```
$ cabal install
```

(v) Testing your installation

```
$ lambdaShell --version  
  
The Lambda Shell, version 0.9.9.1  
Copyright 2005-2011, Robert Dockins
```

8 $\text{\LaTeX}$

Obsolete commands

Last modification: 2025-04-12

Do not use `\bf`, `\em`, `\it`, `\rm`, `\sf`, `\sl` or `\tt`.

The above commands have some issues. For example, how produce bold italic text?

Both `{\bf \it Hello World!}` or `{\it \bf Hello World!}` do not work. You should use `\textbf{\textit{Hello World!}}` or `\textit{\textbf{Hello World!}}` instead.

See, [What's wrong with `\bf`, `\it`, etc.?](#)